

# Template

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9 DAYS · 37 STOPS · GRADE 12 · INSTRUMENTS

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## THE OPENING CARD

This is the format harness rather than a game. Four days hold one stop of every question format the engine can draw, so that a panel can be opened, answered right, answered wrong and looked at, without playing a campaign to the day that happens to hold it. You are whoever is checking that they all still work after a change to the engine. Nothing here teaches a course, and every one of the twenty-eight is authored in the same book format a real game uses.

## Three instruments, one reference

### BEFORE THE DAY — GREET

#### Fourteen people who check each other's work

The office verifies other people's measurements, which means every result here is read by somebody who did not take it. Raman wants you known to all fourteen before Monday, because a query about a number lands very differently from somebody whose name the analyst knows.

### PLAN CARD 112W

Tuesday, and the Halden site has filed a fortnight of readings that all agree with each other, which is what Ada Merrick does not like about them. The response desk wants to publish by noon; Priya Naidoo has found a signature on the closing checklist from somebody who was ashore that week. Today you establish which of the site's numbers were computed from a shared reference, account for a flow nobody has been adding up, and decide which of the closing claims can be signed off on the evidence actually present. A number that inherits somebody else's error is not wrong on its own terms, which is exactly why nobody catches it.

### BRIEFING 13W

Four channels agree, one gauge does not, and the bilge is still rising.

### WHAT YOU DECIDE 16W

Find out whether the numbers agree because they are right or because they share a source.

## 1.1 Four channels, one denominator

FIELD STATION • A ROOM • TRACE

### SCENE 32W

Four of Halden's five numbers agree to within a few percent. The fifth, a mechanical gauge nobody has touched in a year, does not. Merrick has the derivation sheets and is waiting.

### GUIDE 59W

Open each of the five numbers to see what it was computed from. Keep the ones that stand on their own measurement, untick the ones whose chain runs through a shared step, and name that step. Four agreeing and one disagreeing is not four against one: if the four share a denominator, they are one number quoted four times.

### ASK 19W

Everything except the mechanical gauge agrees. Open the derivations, keep what still stands, and name the source at fault.

### BEHIND THE BUTTON 133W

Why agreement is weak evidence. Numbers derived through a common quantity move together whatever that quantity does. The agreement is then a property of the arithmetic rather than of the world, and it is the most convincing form a systematic error takes.

Why the untouched gauge is worth attention. A mechanical gauge with no shared electronics, no shared calibration and no shared denominator is independent by construction. Being the odd one out makes it suspect; being independent makes it valuable. Those are different facts about the same instrument.

What naming the source buys. Once the shared step is identified it can be corrected, and every dependent number moves together in a known direction. Discarding the disagreeing gauge instead leaves the error in place and removes the only thing that could have found it.

### TAKES AS READ

a ratio inherits the error of whatever it was divided by • two displays fed from one sensor are one measurement

### VERDICT 105W

Three of these numbers are ratios, and a ratio is a comparison: it carries the error of whatever sits underneath the line. The vault was assumed to be rock and is not, so every quantity divided by it is wrong by the same factor and they agree with each other beautifully while doing it. That is what makes a common-mode error survive review — nothing looks inconsistent. The correction is arithmetic on exactly those three and nothing else: the gauge measured a pressure directly and the counts came off a separate network, so neither of them ever touched the bad denominator and neither should move.

### TAKEAWAY 15W

Independence is a property of the measurement chain, not of how many displays report it.

## 1.2 What the level rise leaves out

SAMPLE ROOM • A PERSON • BALANCE

### SCENE 33W

The bilge is up eight centimetres in a minute over four square metres of deck. The drain pump has been running the whole time. Tomas Oyelaran wants the leak rate, not the rise.

### GUIDE 54W

Reading a stream costs nothing, so read them all. Then count only what belongs in the total Oyelaran asked for: the water entering at the hull, before the pump takes any out. Eight centimetres over four square metres in a minute is the rise, and the pump has been running for that whole minute.

### ASK 12W

Read what you need and report the total inflow at the hull.

### BEHIND THE BUTTON 127W

Why the rise is not the leak. What accumulates is what came in minus what has been removed. The pump was working throughout the measurement, so its contribution has already been subtracted from what you can see. Adding it back gives the inflow.

How the geometry enters. A depth in centimetres becomes a volume only through the area it sits on, and a volume becomes a rate only through the time it took. Four square metres and one minute are both part of the measurement, not context.

Why the leak rate rather than the level. The level says how bad it is now. The leak rate says whether the pumps can win and how long there is, which is what any decision after this one depends on.

### TAKES AS READ

a volume rate is a level rate multiplied by a plan area • what accumulates is what is left after everything already leaving has left

### VERDICT 100W

The level in the bilge is not the leak; it is what got past the pump. Anything already being removed while you were measuring has to go back into the total, and a pump that has been running since the alarm is easy to leave out precisely because it is not a reading — nothing on the panel changes when it runs. Report the 320 and the response is sized for a leak 2 thirds of its actual rate, which decides how many pumps are worth starting and, more to the point, whether starting pumps is the answer at all.

### TAKEAWAY 13W

An accumulation rate is the inflow minus everything already being removed from it.

## 1.3 The signature and the thing it stands for

RECORDS OFFICE · A ROOM · ATTEST

### SCENE 27W

The closing checklist is signed throughout. Naidoo has the watch bill open beside it and has not said anything yet. There is time to verify three items.

### GUIDE 50W

Open each item on the checklist and read what actually stands behind it. Some are backed by an observation, some by another signature, and some by the watch bill saying somebody was on duty. Verify what the three checks allow, and hold whatever cannot be closed on the evidence present.

### ASK 15W

Verify what the time allows and hold whatever cannot be closed on the evidence present.

### BEHIND THE BUTTON 130W

What a signature records. That somebody took responsibility for a line. Not that the thing was seen, tested or measured. On a closing checklist those two coincide most of the time, which is exactly why the exceptions are hard to notice.

Why the watch bill is beside it. It says who was where. A claim signed by somebody the bill puts elsewhere is not necessarily false, but it is a claim about what they were told rather than about what they saw.

Why holding is the useful outcome. Three checks against a full checklist means most items stay unverified, and that is fine as long as it is recorded. An unverified item marked as such is honest; the same item read later as closed is how a checklist becomes fiction.

### TAKES AS READ

a signature records responsibility, not a physical condition · an indication reports a commanded state rather than a measured one

### VERDICT 101W

A checklist is a compression of experience, and its signatures record who accepted each claim at the time. That is worth having and it is not the same as the condition being true now. The two items that fail here fail differently: one was signed for a space nobody could enter, and one is contradicted by a direct measurement at the boundary, which beats a written record every time because it is the actual state rather than the reported one. Verifying the whole list is not available, so the decision that matters is which claims would hurt if they were wrong.

### TAKEAWAY 16W

A record is evidence that somebody accepted a claim, which is not evidence for the claim.

Agreement between measurements says nothing until you know what they were each computed from.

## What the measurement is worth

### BEFORE THE DAY — TRIAL

#### Walk the floor before the first job

Four areas and a gatehouse, and the work moves between them in a fixed order — station, sample room, records, response desk. Walk it once. The round you will be driving on Friday is this route, and the paperwork order only makes sense once you have seen the distance between the doors.

### PLAN CARD 98W

Wednesday, and Ruth Calderón has one slot left in the day's budget. Three proposals are on her desk and all three are good science; only one of them changes what the crew does at noon. Meanwhile the field station has a tone nobody can source and four candidate machines, and the sample room has a brightness that fits any size you like. Today you decide what to buy, isolate a cause by changing one thing at a time, and find out why a second measurement of different physics settles what a better version of the first one cannot.

### BRIEFING 12W

The desk will fund one more measurement and wants to know which.

### WHAT YOU DECIDE 11W

Spend scarce evidence on the questions that change what happens next.

## 2.1 One slot, three good proposals

RESPONSE DESK · A ROOM · VALUE

### SCENE 26W

Three proposals, one funded slot, and a crew that mobilises at noon either way. Calderón has read all three and will not say which she prefers.

### GUIDE 67W

Two controls set the shot: the angle and the charge. The charge is not calibrated in anything, so change one control at a time and watch where the short mark lands — that is how you find out what each one does. Three marks at three ranges have to be hit, and the same angle will not do for all of them. Put a line on each.

### ASK 17W

You have four hours of the day's budget. Buy what changes what the crew does at noon.

### BEHIND THE BUTTON 181W

Why forty-five degrees is not the answer. Forty-five maximises range for a given launch speed on flat ground with no air. Change the launch height, the drag, or the range you want, and the best angle moves — and for any range short of the maximum there are two angles that reach it, a low fast one and a high slow one, which behave very differently in wind.

Why one control at a time. Two unknown controls changed together give one landing point and no information about either. Fixing the angle and varying the charge, or the reverse, is the same isolation that CONTROL asks for, on a quay with a line thrower instead of a bench.

What the short mark is for. It is the cheapest place to learn: a shot that falls short tells you the sign and roughly the size of your error, and it costs one line to retrieve. Calibrating on the nearest target before attempting the far one is what a gunner does, and it is why the marks are at three ranges rather than one.

### TAKES AS READ

evidence is worth what it changes, not what it costs · a second measurement of the same kind narrows the same thing again

### VERDICT 105W

Two of these proposals measure the surface again. They are careful and they will produce a tighter number about a place nobody is standing, which is the failure this decision exists to avoid: a measurement can be rigorous and still answer a question whose answer does not enter the choice. The test that separates them is whether a result the other way changes anything, and only the gallery reading does — a low oxygen four metres down stops the entry, while a third transect agreeing with the first two stops nothing. The archive is cheap and genuinely informative, and it still does not decide noon.

### TAKEAWAY 15W

A measurement earns its cost by changing a decision that has not yet been taken.



## 2.2 A cause, and a coincidence that looks like one

FIELD STATION • A PERSON • CONTROL

### SCENE 30W

A new tone appeared on the array overnight at 214 hertz. Four machines changed state in the same hour. Achterberg has the plant log and the sonar operator is waiting.

### GUIDE 67W

The number you are watching is the strength of the 214-hertz tone on the array. Four machines changed state in the hour it appeared, which makes four suspects and no evidence. Switching a machine asks whether the tone depends on it; a machine that is innocent leaves the tone where it was. Change one, run the measurement, put it back, then name the machine the tone follows.

### ASK 13W

Four candidates, one tone. Establish which machine is producing it and prove it.

### BEHIND THE BUTTON 151W

Why the log is a list of suspects and not an answer. Four state changes in one hour is ordinary in a working plant, and the tone had to appear during somebody's hour. A record of what changed tells you where to look, and every one of the four would sound equally guilty written down in a report.

What switching actually tests. Stopping a machine removes its contribution and leaves everything else running, so the change in the tone is that machine's own share of it. Two machines stopped together answer a vaguer question — whether either mattered — and that answer will not survive the meeting.

Why the tone comes back. A cause is a thing the effect depends on, in both directions. Restarting the machine and watching the tone return rules out the ordinary alternative: that something else was drifting, settling or being switched somewhere else while you worked.

### TAKES AS READ

a controlled test changes one variable and holds the rest • a machinery-order tone scales with the rate of the machine producing it

### VERDICT 99W

Four things changed in one hour, so the log alone cannot name a cause; it can only name candidates. Securing several at once makes the display quieter and settles nothing, because a quieter display is consistent with any of them being responsible. One variable at a time gives a candidate, and restoring it is what turns the candidate into a cause: a coincidence does not come back on demand. The reversal also costs almost nothing here, which is worth noticing — the reason it gets skipped is not expense, it is that by then the answer already feels known.

### TAKEAWAY 11W

A relationship is causal when it reverses, not when it coincides.

## 2.3 Any size you like, for the right albedo

SAMPLE ROOM • A ROOM • DEGENERACY

### SCENE 28W

One brightness measurement, and a model with two knobs in it. Kit Sorensen has the thermal data on a second screen and has not brought it over yet.

### GUIDE 57W

Two knobs, one brightness measurement, and a family of settings that all match it. Slide along the family and see how wide it is before saying anything about size. The thermal data is on a second screen and has not been brought over — until it is, the honest answer is a range rather than a number.

### ASK 9W

Match the brightness, then say what size it is.

### BEHIND THE BUTTON 119W

Why one measurement cannot fix two unknowns. Brightness is the area times how reflective the surface is. Halve the reflectivity and double the area and the measurement is unchanged. That trade is the family, and it spans a factor of several in diameter.

What the thermal data does. Whatever is not reflected is absorbed and re-emitted as heat, so the infrared brightness depends on the same two quantities in a different combination. Where the two constraints cross is the answer.

Why reporting the family is not a failure. A single number derived from one measurement is a number the data does not contain. Saying what range fits, and what would narrow it, is the professional answer at that moment.

### TAKES AS READ

reflected brightness depends on the product of area and reflectivity • absorbed and reflected fractions of the incident light sum to one

### VERDICT 105W

The brightness is the product of an area and a reflectivity, so halving one and quadrupling the other reproduces it exactly. That is not a noisy measurement; it is a measurement of one number where two are wanted, and a better telescope slides along the same curve. Thermal infrared breaks it because it responds to the absorbed fraction rather than the reflected one — the same albedo that makes an object faint makes it warm — so its curve runs the other way and the two cross once. What comes out is a diameter. The density underneath it is still an assumption wearing a number.

### TAKEAWAY 13W

One observable across two unknowns fixes a family of solutions, not a solution.

### END OF THE DAY 17W

The value of a measurement is the decision it would change, not the precision it would add.

## What the plan cannot answer

### BEFORE THE DAY — FOLLOW

#### The sampling route, walked once by Merrick

Merrick, the station lead, walks the sampling route once a fortnight and does not wait for anybody. Stay inside the band behind her: too far and you lose which way she turned, too close and you are standing in front of what she stopped to look at.

### PLAN CARD 96W

Thursday, and Priya Naidoo has three jobs that have to close before the report does. The gym next to the school is cleared on a rapid inspection that never traced a load path; the river programme has 20 laboratory slots and 6 questions that all deserve them; and the overnight event has 3 arrival times and 1 station whose clock has been drifting since the spring. Today you follow force from a roof to the ground, decide what a sampling plan is allowed not to answer, and report a location as the region it actually is.

### BRIEFING 16W

The gym is cleared on paper, the crate holds 20 slots, and 3 stations disagree politely.

### WHAT YOU DECIDE 19W

Trace a path, spend a finite pool, and locate something from constraints that do not meet at a point.

### 3.1 The small anchor and the large panel

RECORDS OFFICE • A ROOM • CHAIN

#### SCENE 29W

The gym is cleared on a rapid inspection. Naidoo has the drawings out and the roof-to-panel anchors are the only thing on them with no capacity written beside it.

#### GUIDE 61W

Put the transfers in the order the force actually takes, from the moving panel to the ground. Then name the one that decides whether the path holds. The anchors are the only thing on the drawings with no capacity written beside them, which is a fact about the paperwork — the readings on each link are what settle which transfer governs.

#### ASK 19W

Put the transfers in the order the force takes, then name the one that decides whether the path holds.

#### BEHIND THE BUTTON 148W

Why a path is limited by its smallest transfer. Force handed from one element to the next has to be carried by every handover along the way. A large panel and a large roof are irrelevant if the connection between them cannot pass what arrives, and that is nearly always a connection rather than a member.

Why the anchors have no number. Connections are detailed late, sometimes on site, and often by somebody other than the engineer who sized the members. A blank on a drawing is not evidence that the anchor is weak; it is evidence that nobody has checked.

What a rapid inspection can and cannot say. It clears the obvious: no collapse, no visible distress, nothing hanging. It does not open a connection, and this whole stop is about the difference between a building that looks intact and a load path that has been demonstrated.

#### TAKES AS READ

lateral force reaches the ground through a continuous set of physical connections • a chain of transfers is limited by the weakest one the path requires

#### VERDICT 115W

Force does not travel from a roof to the ground by being near it. Every interface has to carry what the one above it delivers, and the path is limited by whichever required transfer is weakest — which has nothing to do with the size of the elements on either side. The panel is the largest thing here by a wide margin and it is not the question; if the anchor cannot pass the roof and wall inertia across the joint, the panel's own strength has nothing to act against and a heavy element is left unsupported out of plane. That is why the item with no number beside it is the item to test.

#### TAKEAWAY 19W

A load path is governed by its weakest required transfer, whatever the elements either side of it can carry.

### 3.2 What 20 bottles cannot ask

RESPONSE DESK · A PERSON · ALLOCATE

#### SCENE 29W

A crate with 20 laboratory slots in it, the river map on the wall, and the partition model's prediction pinned beside it. Nobody has written down a recommended plan.

#### GUIDE 53W

Three proposals, one funded slot, and a crew that mobilises at noon whatever you decide. Open each and ask what the crew would do differently at noon depending on how it comes out. A proposal whose result changes nothing before mobilisation has spent the four hours on something you could have bought tomorrow.

#### ASK 15W

20 slots. Build the plan, and know which of the four questions it leaves open.

#### BEHIND THE BUTTON 125W

What makes a purchase decisive. Two possible results that lead to different actions. Calderón will not say which she prefers because the preference is not the point — the question is which result could change the plan, and that is a property of the proposal rather than of anybody's judgement.

Why the mobilisation time is the constraint. The crew leaves at noon either way, so information arriving afterwards is information about a decision already taken. The whole value of the slot is that it lands before the trucks do.

Why all three are good. They are meant to be. A choice between one good and two poor proposals tests nothing; the skill is separating valuable from decisive when everything on the table is worth doing.

#### TAKES AS READ

a contaminant divides between water, suspended particles and sediment · a measurement without a blank cannot be defended afterwards

#### VERDICT 107W

Every line on this crate is worth sampling, which is exactly why a list of options cannot teach the decision: read one at a time they are all justified. A finite inventory makes the trade concrete — the sediment cores and the second time point together cost more than what is left after the protected slots, so a plan that asks whether the concentration is rising cannot also ask whether there is a reservoir underneath it. The error worth avoiding is not choosing badly. It is finishing the programme without having noticed that a question was silently dropped, and reporting reassurance built on the phase nobody sampled.

#### TAKEAWAY 14W

Sampling is experimental design under a budget, and every feasible plan forgoes a question.

### 3.3 Three circles and an area

FIELD STATION · A ROOM · TRIANGULATE

#### SCENE 30W

Three stations, three arrival separations, and a regional map. The ridge station's clock has been drifting since the spring and there is a correction sheet for it on the bench.

#### GUIDE 75W

Switching a station in draws the region its own measurement permits — a ring, not a point, because an arrival separation fixes a distance and says nothing about direction. Two rings cross in two places; three settle it. One station's clock has been drifting, and there is a correction sheet for it on the bench, so decide what to do about that before you commit. Then click the map where you think the source is.

#### ASK 16W

Switch the stations in, decide what to do about the ridge clock, and report the position.

#### BEHIND THE BUTTON 172W

Why one station gives a ring. What is measured is when the signal arrived relative to another station, which fixes how much further it travelled — a distance, with no information about bearing. Every point at that distance is equally consistent, and that set of points is the ring you see drawn.

Why the third station matters. Two rings intersect in two places, and both are consistent with everything measured so far. A third ring picks one of them, and the size of the region where all three overlap is the position uncertainty. That region, not its centre, is what a report should describe.

What a systematic does, and why it is not noise. A drifting clock offsets one station's arrival time by the same amount every time, so its ring is drawn in the wrong place — consistently. More measurements from that station will not average it away; only the correction will. That is the difference between an error you can beat with statistics and one you have to know about.

#### TAKES AS READ

two wave types travelling at different speeds separate further with distance · a location computed from arrival times inherits any error in those times

#### VERDICT 98W

One station's arrival separation says how far the source was. It says nothing about direction, so on its own it leaves a whole ring. Three rings intersect, and what they intersect in is an area, not a point. Reporting a coordinate claims a precision the arrival times do not support, and somebody downstream will treat it as certain. The clock matters because the calculation runs on times, not on waveforms. A station biased by six tenths of a second draws its ring in the wrong place. Its own record looks perfectly normal, and the intersection moves with it.

#### TAKEAWAY 17W

One station gives a distance, three give a region, and a timing error moves the whole region.

### 3.4 What the level rise leaves out

SAMPLE ROOM • A ROOM • CALLBACK • BALANCE

#### SCENE 33W

The bilge is up eight centimetres in a minute over four square metres of deck. The drain pump has been running the whole time. Tomas Oyelaran wants the leak rate, not the rise.

#### GUIDE 54W

Reading a stream costs nothing, so read them all. Then count only what belongs in the total Oyelaran asked for: the water entering at the hull, before the pump takes any out. Eight centimetres over four square metres in a minute is the rise, and the pump has been running for that whole minute.

#### ASK 12W

Read what you need and report the total inflow at the hull.

#### BEHIND THE BUTTON 127W

Why the rise is not the leak. What accumulates is what came in minus what has been removed. The pump was working throughout the measurement, so its contribution has already been subtracted from what you can see. Adding it back gives the inflow.

How the geometry enters. A depth in centimetres becomes a volume only through the area it sits on, and a volume becomes a rate only through the time it took. Four square metres and one minute are both part of the measurement, not context.

Why the leak rate rather than the level. The level says how bad it is now. The leak rate says whether the pumps can win and how long there is, which is what any decision after this one depends on.

#### TAKES AS READ

a volume rate is a level rate multiplied by a plan area • what accumulates is what is left after everything already leaving has left

#### VERDICT 100W

The level in the bilge is not the leak; it is what got past the pump. Anything already being removed while you were measuring has to go back into the total, and a pump that has been running since the alarm is easy to leave out precisely because it is not a reading — nothing on the panel changes when it runs. Report the 320 and the response is sized for a leak 2 thirds of its actual rate, which decides how many pumps are worth starting and, more to the point, whether starting pumps is the answer at all.

#### TAKEAWAY 13W

An accumulation rate is the inflow minus everything already being removed from it.

END OF THE DAY 16W

A finite plan is defined as much by what it forgoes as by what it covers.



## Before the number moves

### PLAN CARD 97W

Friday, and three decisions have to be made in the order that makes them possible. The entry solution is a distribution and the corridor is two and a half degrees wide; the deflection was delivered on Tuesday and nobody has yet said what it did; and the public-response thresholds have to exist before tonight's orbit update rather than after it. Calderón will hold you to whichever rules you write. Today you narrow a spread instead of moving it, spend the tracking to find out what actually happened, and set thresholds against lead times that cannot be compressed.

### BRIEFING 12W

Calderón wants the response board signed before tonight's update, not after it.

### WHAT YOU DECIDE 18W

Judge a spread against a limit, measure what you changed, and write the rules before the data arrives.

## 4.1 Moving the dot and moving the cloud

SAMPLE ROOM · A ROOM · CLOUD

### SCENE 29W

The corridor runs 5.3 to 7.7 degrees. The current solution sits at 6.9 with 9 tenths of a degree on it, and a fifth of the trajectories are outside.

### GUIDE 71W

You drive the round yourself. Every gate is lit and none is marked as next, because the order is the answer: work out which step has to come before which, and drive them in that sequence. The clock is only a clock — what is graded is the order you took, not how fast you took it. Take the gates in the order the sample actually has to move through them.

### ASK 24W

Bring the entry solution inside the corridor, place the bars to report the middle and its one-sigma uncertainty, and say when it is ready.

### BEHIND THE BUTTON 164W

Why an order exists at all. Each station either changes the sample or measures it, and a measurement is only meaningful on a sample that has had everything before it done and nothing after it. Weigh it after drying and the number means one thing; weigh it before and it means another. So the sequence is not a convention — it is what makes each number interpretable.

Why nothing marks the next gate. A route that lights up in order tests whether you can follow instructions. This one tests whether you know why the order is what it is, which is the thing that survives a day when the instructions are missing or wrong.

Why the clock is not the grade. Driving fast in the wrong order arrives with the work undone, and a laboratory that rewards speed over sequence produces numbers nobody can use. The time is there because a round has to fit in an afternoon, not because it is being scored.

### TAKES AS READ

a state estimate is a distribution over what the measurements permit · the limits of a corridor come from survivability, not from navigation

### VERDICT 86W

The nominal angle is the one place the vehicle is least likely to be. Re-targeting it slides the whole distribution across without changing its width. The fraction outside the corridor barely moves; the trajectories that were long become shallow instead. What is being asked is not where to aim but how much the measurements permit, and only new tracking reduces that. It costs hours the day would rather spend elsewhere. That is the real decision, because the corridor is a survivability limit and does not negotiate.

### TAKEAWAY 11W

Re-centring moves the whole distribution; only new information makes it narrower.

## 4.2 A confirmed impact is not a confirmed deflection

RECORDS OFFICE · A PERSON · VERIFY

### SCENE 30W

The impactor arrived on Tuesday and the confirmation is on the wall. The ejecta enhancement was quoted as somewhere between one and five. Three tracking passes remain in the budget.

### GUIDE 56W

Lock a predicted period change before committing the impact, and it cannot be edited afterwards. Then run it, then spend one of the three tracking passes to find out what actually happened. The enhancement was quoted as somewhere between one and five, so your prediction is a range with an assumption in it — say which.

### ASK 11W

Predict the period change, commit the impact, and report what happened.

### BEHIND THE BUTTON 137W

Why a confirmed impact is not a confirmed deflection. Arriving is a fact about the spacecraft. Moving the target is a fact about momentum, and most of the momentum transfer comes from material thrown off the surface, which nobody has measured for this body.

What an enhancement between one and five means. One means the impactor's own momentum and nothing more. Five means the ejecta carried four times as much again. That range is the difference between a demonstration and a technique, and only the tracking pass can narrow it.

Why the prediction is locked. With the measurement in hand, an enhancement can be chosen to fit and the prediction will appear to have been right. Committing to the range first is what makes the pass capable of measuring the one quantity the mission existed to measure.

### TAKES AS READ

momentum leaving a surface as ejecta carries some of the transferred impulse · an orbital period change is measurable by tracking

### VERDICT 107W

The spacecraft's own momentum is known to a few percent. What is not known is how much more the ejecta carried away, and quoting that as a single number rather than the factor of several it actually is turns an honest uncertainty into a false prediction. Both halves of the error are ordinary; the failure that is specific to this decision is finishing without measuring. Confirmation of impact is confirmation of an event, not of an effect, and if the tracking budget goes on refining the pre-impact orbit there is nothing left to compare it against — the campaign then produces a photograph instead of a result.

### TAKEAWAY 15W

An intervention is verified by measuring what it changed, not by confirming that it happened.

### 4.3 Rules, and the lead time they need

RESPONSE DESK · A ROOM · TRIGGER

#### SCENE 30W

The response board is blank. The current probability is one in two and a half thousand, the next update lands tonight, and Calderón wants the board signed before it does.

#### GUIDE 65W

One rule, set on the probability scale and signed before tonight's update exists. The evacuation takes fourteen days from the moment it is ordered, and only the early updates arrive with that much time left. Set the line low enough that one of them crosses it, and high enough that it would not have been crossed by an object which turned out to be nothing.

#### ASK 15W

At what impact probability do you begin the staged evacuation, given it needs fourteen days?

#### BEHIND THE BUTTON 207W

Why the board is signed first. A threshold moved while the updates come in is not a threshold, it is a description of what somebody has already decided. The signature is what makes tonight's actions attributable to a rule rather than to whoever happened to be at the desk.

What lead time is, and why every stage has a different one. An advisory can be published in an afternoon; an evacuation order cannot. The lead time is how long the action takes to arrive once it is called, so a stage set close to the end of the campaign starts something that cannot finish — worse than no rule, because it is a rule somebody trusted.

Why a line can also be drawn too early. A stage that fires on a reading which had shown nothing is a standing order, not a rule: it would have fired on any campaign whatever the object did. Both failures are graded, and they look nothing alike from the desk.

What the schedule is for. The updates arrive on a known timetable even though their contents are unknown, so the real decision is which update you want each stage to act on — and then the number that puts it there.

#### TAKES AS READ

a probability is a statement about knowledge rather than about the object · a staged response has a lead time that cannot be compressed

#### VERDICT 100W

Two ways of getting this wrong look nothing alike and cost the same. A threshold set too high never fires, and the stage simply does not happen. A threshold set at a sensible number fires on the day 9 update with 300 hours left, which is a correct rule for an action that needed 336 — so the rule is satisfied and the evacuation is already impossible. The day 15 collapse to 1 in 9000 does not make the day 9 decision a

mistake, and the board is what makes that arguable afterwards, because it was signed before anybody knew.

**TAKEAWAY 17W**

A rule written before the data is a decision; the same rule written afterwards is a reaction.

## 4.4 Four channels, one denominator

FIELD STATION · A ROOM · CALLBACK · TRACE

### SCENE 32W

Four of Halden's five numbers agree to within a few percent. The fifth, a mechanical gauge nobody has touched in a year, does not. Merrick has the derivation sheets and is waiting.

### GUIDE 59W

Open each of the five numbers to see what it was computed from. Keep the ones that stand on their own measurement, untick the ones whose chain runs through a shared step, and name that step. Four agreeing and one disagreeing is not four against one: if the four share a denominator, they are one number quoted four times.

### ASK 19W

Everything except the mechanical gauge agrees. Open the derivations, keep what still stands, and name the source at fault.

### BEHIND THE BUTTON 133W

Why agreement is weak evidence. Numbers derived through a common quantity move together whatever that quantity does. The agreement is then a property of the arithmetic rather than of the world, and it is the most convincing form a systematic error takes.

Why the untouched gauge is worth attention. A mechanical gauge with no shared electronics, no shared calibration and no shared denominator is independent by construction. Being the odd one out makes it suspect; being independent makes it valuable. Those are different facts about the same instrument.

What naming the source buys. Once the shared step is identified it can be corrected, and every dependent number moves together in a known direction. Discarding the disagreeing gauge instead leaves the error in place and removes the only thing that could have found it.

### TAKES AS READ

a ratio inherits the error of whatever it was divided by · two displays fed from one sensor are one measurement

### VERDICT 105W

Three of these numbers are ratios, and a ratio is a comparison: it carries the error of whatever sits underneath the line. The vault was assumed to be rock and is not, so every quantity divided by it is wrong by the same factor and they agree with each other beautifully while doing it. That is what makes a common-mode error survive review — nothing looks inconsistent. The correction is arithmetic on exactly those three and nothing else: the gauge measured a pressure directly and the counts came off a separate network, so neither of them ever touched the bad denominator and neither should move.

### TAKEAWAY 15W

Independence is a property of the measurement chain, not of how many displays report it.

#### END OF THE DAY 20W

A rule written after the number moves is not a rule, and an action nobody measured is not a result.

## Which number is carrying the error

### BEFORE THE DAY — HUNT

#### Four reference blocks, and the store lists two

Every calibration in this office traces back to a reference block, and two of them are not where the register says. A block nobody can find is a chain of measurements nobody can defend, which is the one thing this office exists not to have.

### PLAN CARD 98W

Monday, and Ada Merrick has an energy figure quoted to three significant figures that nobody can defend. The radius is measured, the density is inferred, and the two are being treated as though they were the same kind of number. Meanwhile the remediation board has four options that are only ranked at one value of a containment endurance nobody measured, and the subsidence survey has two fits with a tenth of a millimetre between them. Today you find which input carries the uncertainty, which option survives the assumption moving, and which fit you would be willing to propagate.

### BRIEFING 13W

Three jobs where the obvious quantity is not the one that decides anything.

### WHAT YOU DECIDE 12W

Find what actually dominates an uncertainty, and what survives an assumption moving.



## 5.1 The exponent is not the whole story

FIELD STATION • A ROOM • PROPAGATE

### SCENE 38W

The energy figure is on the board to three significant figures. Merrick has the three inputs it was built from beside it, each with what it is actually known to, and one measurement slot left in the day.

### GUIDE 76W

The number to hold is the room's temperature, and the trim heater is the only thing you can push with. Nothing tells you which way it pushes or how hard, so find that out early — while the band you have to stay inside is still wide. It narrows as the run goes on, and the disturbances arrive whether or not you are ready. Keep the room inside the band for the quarter of an hour.

### ASK 14W

One measurement is affordable. Which of these is carrying the uncertainty in the energy?

### BEHIND THE BUTTON 171W

Why the band narrows. A mass comparison gets more sensitive as it settles, so the same temperature error matters more at the end than at the beginning. The band is the tolerance the comparison can absorb at that moment, which is why a run can be comfortable for ten minutes and unrecoverable in the last two.

Why you should learn the control before you need it. A heater has a delay: it changes the rate at which the room warms, not the temperature itself, so a correction applied when you are already outside the band arrives late. Nudging it once early, and watching what happens, converts an unknown control into a known one at the cheapest moment.

What the propped door is doing there. Every disturbance in this run is something ordinary in a working building — a door, a lamp, somebody in the corridor. None of them announce themselves, and none of them wait. Holding a quantity steady in a room full of other people's work is the actual job.

### TAKES AS READ

fractional uncertainties combine weighted by the power each term is raised to • a measured quantity and an inferred one are different kinds of number

### VERDICT 100W

A term contributes its exponent multiplied by its own fractional width. The radius is cubed, which is the part everybody sees, and it is known well enough that 3 times 8% is still smaller than 1 times 45. Buying another radar pass therefore improves the number that was already good and moves the total almost not at all. What makes this worth catching is that the shortcut — rank by exponent — is reasonable, teachable and produces exactly the wrong answer here. The density is also the term nobody can measure directly from orbit, which is why it stays wide.

### TAKEAWAY 26W

A term's contribution is its own width times its exponent, so a badly known number with no power over it can dominate a well-known one cubed.

## 5.2 Best at the nominal, and nowhere else

SAMPLE ROOM • A PERSON • STRESS

### SCENE 34W

Four remediation options are ranked on the board, all at one figure for how long the containment holds. Tomas Oyelaran has the range that figure actually came with and has said nothing about it.

### GUIDE 71W

The band is every entry angle the tracking permits — a fifth of it currently outside the corridor. Four actions are available, and they do two different kinds of thing to the band: some move it, some change how wide it is. Try them and watch the centre and the width separately. Then say when the solution is ready, which is a claim about the whole band rather than its middle.

### ASK 14W

Move the endurance through its range, then choose the option you would commit to.

### BEHIND THE BUTTON 158W

What the band is, exactly. It is not a probability of being right about one number: it is the set of trajectories the measurements cannot rule out. Everything inside it is a trajectory this vehicle might actually fly, which is why a criterion is written as a fraction of the band inside the corridor rather than as a best estimate.

Two kinds of action. Retargeting and trimming move where the band sits. A tracking pass adds information and narrows it. A band whose width is wider than the corridor cannot be rescued by moving it, however good the centre looks — and a display that reports only the centre will show that rescue happening.

Why readiness is part of the answer. Actions cost, and there is no configuration in which the band is a point. So the decision has two halves: which work to buy, and when what you have is inside the corridor by enough to commit.

### TAKES AS READ

an assumption stated as a range is not a value • an option is only available while the conditions it needs still hold

### VERDICT 98W

At 78 hours the direct excavation finishes first and looks like the obvious call. The endurance was never 78 hours; it was a range with 78 in the middle of it, and at the low end the direct option needs more containment than there is. What the slider shows is that three of the four options are conditional on an assumption holding, and the ranking everybody was reading is a ranking of what happens if it does. Choosing the staged option costs 13 hours and buys the property that the plan does not have to be abandoned partway.

### TAKEAWAY 17W

A choice that is only best at the nominal is a choice that has not been tested.

### 5.3 A pattern is not an outlier

RECORDS OFFICE • A ROOM • RESIDUAL

#### SCENE 34W

Three fits of the same subsidence survey are on Priya Naidoo's screen, within two hundredths of a millimetre of each other on the summary statistic. The residuals are plotted where each marker actually sits.

#### GUIDE 53W

Three fits, and their summary statistics sit within two hundredths of a millimetre of each other. That number cannot choose between them. The residual field can: each stick is one marker, drawn where it actually sits. Look at all three before choosing, and propagate the fit whose leftovers look like nothing in particular.

#### ASK 15W

Look at each field before you choose. Which of these fits would you propagate forward?

#### BEHIND THE BUTTON 148W

Why the summary statistic cannot decide. It adds up how large the misses are and discards where they were. Two fits can miss by the same total, one scattering its misses at random and the other getting a whole region wrong in the same direction — and only the second is missing a term.

What structure in residuals means. Leftovers that rise across a corner of the survey, or that all lean one way near a boundary, say the model has not accounted for something real. Noise is what remains when the model is complete; a pattern is a message about what it lacks.

Why the lowest number is a trap. Fits are compared by statistic because it tabulates neatly, and the most flexible model usually wins on it while performing worst on next season's data. Choosing on the field is choosing on whether the model is right.

#### TAKES AS READ

a complete model leaves residuals that look like noise • a fitted parameter can absorb a systematic without removing it

#### VERDICT 107W

Two of these three leave residuals that are positive on one side of the site and negative on the other. That is not scatter; it is a quantity the model does not contain, and adding a free parameter that soaks it up improves the summary statistic while leaving the cause in place. It will then bias every extrapolation in the same direction, quietly, because nothing about the fit looks wrong afterwards. The re-levelled fit scores worse and its residuals change sign at random, which is what a complete model produces. One large residual would be a bad marker; six small ones in a pattern are a cause.

#### TAKEAWAY 21W

A low summary statistic with a pattern under it hides a systematic that will bias every later prediction the same way.

## 5.4 One slot, three good proposals

RESPONSE DESK · A ROOM · CALLBACK · VALUE

### SCENE 26W

Three proposals, one funded slot, and a crew that mobilises at noon either way. Calderón has read all three and will not say which she prefers.

### GUIDE 67W

Two controls set the shot: the angle and the charge. The charge is not calibrated in anything, so change one control at a time and watch where the short mark lands — that is how you find out what each one does. Three marks at three ranges have to be hit, and the same angle will not do for all of them. Put a line on each.

### ASK 17W

You have four hours of the day's budget. Buy what changes what the crew does at noon.

### BEHIND THE BUTTON 181W

Why forty-five degrees is not the answer. Forty-five maximises range for a given launch speed on flat ground with no air. Change the launch height, the drag, or the range you want, and the best angle moves — and for any range short of the maximum there are two angles that reach it, a low fast one and a high slow one, which behave very differently in wind.

Why one control at a time. Two unknown controls changed together give one landing point and no information about either. Fixing the angle and varying the charge, or the reverse, is the same isolation that CONTROL asks for, on a quay with a line thrower instead of a bench.

What the short mark is for. It is the cheapest place to learn: a shot that falls short tells you the sign and roughly the size of your error, and it costs one line to retrieve. Calibrating on the nearest target before attempting the far one is what a gunner does, and it is why the marks are at three ranges rather than one.

### TAKES AS READ

evidence is worth what it changes, not what it costs · a second measurement of the same kind narrows the same thing again

### VERDICT 105W

Two of these proposals measure the surface again. They are careful and they will produce a tighter number about a place nobody is standing, which is the failure this decision exists to avoid: a measurement can be rigorous and still answer a question whose answer does not enter the choice. The test that separates them is whether a result the other way changes anything, and only the gallery reading does — a low oxygen four metres down stops the entry, while a third transect agreeing with the first two stops nothing. The archive is cheap and genuinely informative, and it still does not decide noon.

### TAKEAWAY 15W

A measurement earns its cost by changing a decision that has not yet been taken.

END OF THE DAY 14W

A quantity is only as good as the widest thing it was built from.

## What you take yourself

### BEFORE THE DAY — CANVASS

#### Which analysts actually saw the revised method

The method was revised three weeks ago and the office assumes everybody is using it. Ask round the areas until enough answers have come back to say whether the results this month are on one method or two.

### PLAN CARD 98W

Tuesday, and four conditions arrive on Ruth Calderón's desk inside a minute. One of them is loud and none of the loud ones is the one with a margin running out. In the flooded gallery the survey trolley runs on a rail with nothing to slow it down, and the last operator stopped the motor at the mark and watched it carry on past. Kit Sorensen has the injection results for the monitoring network. Today you decide what you keep, what you hand over and with what condition, and what a network is failing to see at all.

### BRIEFING 18W

Four conditions at once, a trolley that will not stop, and a network that never sees one direction.

### WHAT YOU DECIDE 17W

Decide what command changes first, fly a plan with no damping, and measure your own blind spot.

## 6.1 The loudest is not the first

RESPONSE DESK · A ROOM · DELEGATE

### SCENE 31W

Four conditions, one watch team, and a clock that runs whether or not anybody has decided. Ruth Calderón will take whichever one you leave her, and she wants the others owned.

### GUIDE 55W

Write each stage's threshold on the board before releasing it, and once released the numbers cannot be moved. The next update lands tonight, and every action a stage starts takes time to finish. Set the thresholds so each action completes before the situation it was written for arrives, then stand by what the board does.

### ASK 10W

Which do you change first, and who has the others?

### BEHIND THE BUTTON 143W

Why a probability of one in two and a half thousand is not the trigger. It is the current state, and it will move with tonight's update. Thresholds are about what you will do at values you have not seen yet, which is exactly why they can be set calmly now and not later.

What lead time does to a threshold. An action that takes two weeks to complete must start at a probability that will still be two weeks early. Setting the trigger at the level where the action is needed is setting it at the moment it should already have been finished.

Why signing before the update. A threshold adjusted after seeing a number is not a rule; it is a justification. Calderón wants the board signed tonight because that is the difference between a policy and a series of reactions.

### TAKES AS READ

a trend and a rate say more than a current value · an instruction without a return condition produces no report until it is too late

### VERDICT 110W

The fire panel is the loudest thing in the building and it has been stable for 20 minutes on a fault that is already in the log. The spill is rising and it is inside a bund, so what it costs is money and time rather than an option. The groundwater is the only condition where waiting removes a choice: over the sill, the records are gone and no amount of later effort brings them back. That is what puts it first — not the rate, not the noise, but a margin closing on something irreversible. Everything else can be somebody else's if they are told when to come back.

### TAKEAWAY 23W

A handover is an owner, a first action and a condition that brings them back; two of the three is nobody watching it.



## 6.2 You stopped driving. You did not stop

FIELD STATION · A PERSON · FLY

### SCENE 38W

The survey trolley runs on a rail in the flooded gallery, where nothing slows it down. Ben Achterberg has the last operator's trace, which shows the motor cut at the mark and the trolley 18 metres past it.

### GUIDE 59W

The panel shows an error budget: each input's own width, multiplied by the power it carries in the formula, contributes a bar. Read the bars rather than the inputs. One measurement is affordable, so buy the one that shortens the tallest bar — a large exponent on a well-measured quantity can matter less than a modest one measured badly.

### ASK 18W

Set the pulse and the distance at which braking starts, then run it and see where it stops.

### BEHIND THE BUTTON 128W

How the widths combine. For a product of powers, the fractional uncertainties add in quadrature, each weighted by its exponent. That is why the answer is a sum of contributions rather than 'the worst input', and why the bars are the honest display.

Why three significant figures is a claim. Quoting an energy to three figures asserts a fractional uncertainty below about a per cent. If no combination of the inputs supports that, the figure is a statement the measurement cannot back — before anybody argues about which input to improve.

What to do if the tallest bar cannot be improved. Say so, and quote fewer figures. A budget that identifies an irreducible term has still done its job: it turns an indefensible number into a defensible one.

### TAKES AS READ

a body with no resisting force keeps whatever velocity it was given · removing a velocity takes the same impulse that produced it

### VERDICT 113W

The trolley leaves the acceleration with a speed, and on a rail in still water there is nothing to take that speed away except an equal impulse in the other direction. So the braking has to start early, and by a specific amount: the distance covered while the brake is working, which is set by the same pulse that produced the speed.

Cutting the drive at the mark is not braking at all — it only stops adding speed, and the trolley carries the speed it already has. A longer pulse gets there sooner and needs a longer brake and more of the budget, which is the trade the two controls actually make.

### TAKEAWAY 23W

With nothing to damp it, the braking impulse has to begin before the target, by exactly as far as the acceleration carried it.

## 6.3 More detections, the same amount of notice

SAMPLE ROOM • A ROOM • INJECT

### SCENE 34W

5000 synthetic events with known times and sizes, ready to push through the monitoring network exactly as a real 1 would go. Kit Sorensen has the recovery display up and nothing on it yet.

### GUIDE 49W

Move the endurance figure through the range it actually came with, and watch which options go dark. All four are ranked at one number, which is why they look separable. Commit to the one that still works at the pessimistic end, not the one that wins in the middle.

### ASK 13W

Run the configurations you want, then fund the upgrade that actually buys something.

### BEHIND THE BUTTON 130W

Why a single figure hides the decision. Ranking four options at one endurance value produces a clean ordering and conceals that the ordering changes across the range. An option best at the nominal and worst at the low end is not the best option; it is the most fragile one.

What the range represents. Not disagreement between people — the honest spread of what is known. Choosing at its middle accepts a real chance of being on the wrong side, and containment failures are not the kind of error that can be corrected afterwards.

Why Oyelaran has said nothing. The board as presented is defensible and incomplete, which is the ordinary way a decision gets made badly. Being the person who asks for the range is most of the job.

### TAKES AS READ

a detection pipeline has a selection function, and it can be measured • a warning is only useful if it arrives while something can still be done

### VERDICT 112W

Lowering the threshold more than doubles the count, which is the number everybody reports, and it adds two hours of notice. What the injection shows is that the extra events are ones the network was already going to catch, only slightly earlier. The direction above the divide produces no recoveries in any configuration at all, and that is not a property of the events — the truth is known and they are there. It is a property of where the stations are. A discovery is worth what the downstream response can do with the time it buys, so a configuration that finds more and warns no earlier has bought a longer list.

### TAKEAWAY 22W

Injecting a population whose truth you know measures the pipeline rather than the world, and what never comes back is not absent.

## 6.4 The signature and the thing it stands for

RECORDS OFFICE · A ROOM · CALLBACK · ATTEST

### SCENE 27W

The closing checklist is signed throughout. Naidoo has the watch bill open beside it and has not said anything yet. There is time to verify three items.

### GUIDE 50W

Open each item on the checklist and read what actually stands behind it. Some are backed by an observation, some by another signature, and some by the watch bill saying somebody was on duty. Verify what the three checks allow, and hold whatever cannot be closed on the evidence present.

### ASK 15W

Verify what the time allows and hold whatever cannot be closed on the evidence present.

### BEHIND THE BUTTON 130W

What a signature records. That somebody took responsibility for a line. Not that the thing was seen, tested or measured. On a closing checklist those two coincide most of the time, which is exactly why the exceptions are hard to notice.

Why the watch bill is beside it. It says who was where. A claim signed by somebody the bill puts elsewhere is not necessarily false, but it is a claim about what they were told rather than about what they saw.

Why holding is the useful outcome. Three checks against a full checklist means most items stay unverified, and that is fine as long as it is recorded. An unverified item marked as such is honest; the same item read later as closed is how a checklist becomes fiction.

### TAKES AS READ

a signature records responsibility, not a physical condition · an indication reports a commanded state rather than a measured one

### VERDICT 101W

A checklist is a compression of experience, and its signatures record who accepted each claim at the time. That is worth having and it is not the same as the condition being true now. The two items that fail here fail differently: one was signed for a space nobody could enter, and one is contradicted by a direct measurement at the boundary, which beats a written record every time because it is the actual state rather than the reported one. Verifying the whole list is not available, so the decision that matters is which claims would hurt if they were wrong.

### TAKEAWAY 16W

A record is evidence that somebody accepted a claim, which is not evidence for the claim.

What decides an order is a margin closing on something irreversible, not which alarm is loudest.

## Down there without the lights

### PLAN CARD 90W

Wednesday, and Priya Naidoo needs the lower vault surveyed before the transfer on Friday. The lighting circuit there has failed twice this fortnight and the second time somebody counted paces to get out, took a detour round a jammed door and finished the count in the wrong room. Ada Merrick wants the route learned as places rather than as a number. Today you walk it once with the labels showing, walk it again without them, and find out what a compartment sequence gives you that a pace count does not.

### BRIEFING 15W

The vault lighting fails on Thursdays and the survey has to be finished either way.

### WHAT YOU DECIDE 13W

Learn a route well enough to rejoin it, and close the fortnight out.

## 7.1 Where you are, and how you know

RECORDS OFFICE • A ROOM • ROUTE

### SCENE 34W

The lower vault runs as one line of rooms with a door at each end. Priya Naidoo walks it with you once with the labels up, and mentions that one of the doors sticks.

### GUIDE 77W

Walk the route once with the labels showing. Then rebuild it from memory, in order, with the labels gone. Partway through, a door is jammed and you come out somewhere the route did not take you, and you have to say which room you are in. Nothing is timed. What is graded is the order and the room you name, and the room is nameable only from what is in it, because the labels are not there.

### ASK 22W

Walk it lit, then put it back together in the dark — and say where you are when the route is broken.

### BEHIND THE BUTTON 207W

Why the count is the trap. Paces and turns both give a number, and a number feels like knowing where you are. But a count cannot be resumed: once the route is broken, the count describes a journey that did not happen, and there is no way to check it from inside the building. It fails silently, which is worse than failing loudly.

Why a list of places survives. Every room on the route can be recognised on its own, from something in it that no other room has. That means the route is re-enterable at any point: being put down in the middle of it costs you the rooms you missed and nothing else. A sequence of recognisable things has that property; a running total never does.

What to look at on the way out. On the lit walk, the labels are the easy thing to read and the useless thing to remember, because they will be gone. What is worth carrying is the one feature of each room you would still know it by in the dark — the rail, the drawers, the smell, the ramp. That is the whole of the method, and it is why the walk is offered before it is needed.

### TAKES AS READ

a count of paces or turns cannot be resumed once it is interrupted • a place can be recognised from something in it that is not a label

### VERDICT 101W

A pace count and a turn count both produce an answer and neither can be resumed. Once the route is broken the count is measuring a journey that did not happen, and there is no way to check it from inside the building. A sequence of places has the property the count does not: every element of it is recognisable on its own, so being put down in the middle of it is recoverable. Naming where you are is the whole of the recovery, and it is only possible because each room has something in it that no other room has.

### TAKEAWAY 16W

A sequence of recognisable places can be re-entered at any point; a count of anything cannot.

## 7.2 What can be signed off

RESPONSE DESK · A PERSON · CHOICE

### SCENE 29W

Ruth Calderón has the fortnight's findings on one sheet and a transfer on Friday. She wants to know which of them she is entitled to put her name to.

### GUIDE 59W

Four sentences, and the office corrected one reference factor and recomputed what divided by it. Ask of each option how far it travels from that. One stretches a single number across a frequency range nobody worked in. One inverts what was found about the independent instrument. And one throws away the correctly measured numerators along with the bad denominator.

### ASK 9W

Which of these can the office state without qualification?

### BEHIND THE BUTTON 195W

Why the wrong options are the interesting ones. Each distractor is written to be the answer under one specific misreading — a step skipped, a quantity confused with a rate, a correlation taken for a cause. Identifying which misreading each belongs to is where the learning is; the right answer alone can be reached on instinct and teach nothing.

Why the shape of an option tells you nothing. A correct answer collects the qualifying clause and the unit. So across a whole game the key used to be the longest option far more often than chance — 88 per cent of passage quizzes here, before it was fixed. It is now checked per game as a binomial tail, so length, hedging and specificity have had the signal taken out of them deliberately.

What the verdict adds. Every wrong option carries its own rebuttal, and the verdict prints them — the reason that option fails, not a restatement of the right one. They are worth reading after a correct answer as well: a right choice made for an approximate reason is indistinguishable from a sound one until the day the approximation is what is being tested.

### TAKES AS READ

a corrected ratio and an uncorrected one are different claims

### VERDICT 83W

The office measured a reference factor and recomputed the ratios that used it. That is a bounded claim with a stated basis, and it is the only one on this list. The second is the same number stretched into a frequency range nobody worked in. The third inverts what was found — the gauge was the independent instrument and it stands. The fourth throws away the correctly measured numerators along with the mislabelled denominator, which is the failure the whole fortnight was about.

### TAKEAWAY 12W

What a verification office establishes is bounded by what it actually measured.



### 7.3 What each claim is standing on

SAMPLE ROOM • A ROOM • PROTOCOL

#### SCENE 22W

Four findings from the fortnight, and four different kinds of evidence underneath them. Tomas Oyelaran wants them paired before anything is filed.

#### GUIDE 56W

Four findings and four kinds of evidence. Pair them by asking what makes each one strong. A direct measurement? One variable changed and put back? Two instruments answering through different physics? Or a record contradicted by an inspection? All four are strong. Confusing which is which is how a bounded claim becomes a general one downstream.

#### ASK 10W

Match each finding to the evidence it actually rests on.

#### BEHIND THE BUTTON 192W

Why this is a matching board and not four separate questions. Responses on boards like this are written to be plausible for more than one situation, so choosing them one at a time lets a nearly-right answer through unexamined. Having to place all of them forces the comparison — not "is this reasonable here", but "is it more right here than there".

How to use the one-each rule. Because the responses are a set to be distributed rather than a list to be sampled, every join you make constrains the rest. Settling the two you are confident of can leave the remaining pair decided by elimination. Where it does not, two responses are still competing for one situation, and that is exactly the distinction the stop is testing.

Why you cannot be wrong about exactly one. With every response used once, three correct joins leave the fourth with only one place to go. Any error therefore involves at least two joins. That is worth remembering when the last pair looks forced: the fault is usually not in the join you are struggling with, but in one you made early and stopped questioning.

#### TAKES AS READ

different kinds of evidence support different kinds of claim

#### VERDICT 28W

Each of these findings is strong, and each is strong for a different reason. Confusing them is how a bounded claim turns into a general one somewhere downstream.

#### TAKEAWAY 20W

A claim is only as strong as the kind of evidence under it, and the two are easy to swap.

## 7.4 What the gate is doing to the river

FIELD STATION · A ROOM · DERIVE

### SCENE 40W

The gate is winding open at a steady rate and the head above the sill is 2.0 metres. Ada Merrick needs the discharge rate written down before the shift changes, and the gauge downstream only reports what has already arrived.

### GUIDE 58W

You set two things: the size of the pulse and the distance at which braking starts. Then the run happens and you watch it. Nothing on this rail slows the trolley down, so cutting the motor is not stopping — whatever you put in has to be taken out deliberately, and the braking has to lead the mark.

### ASK 16W

Derive how fast the discharge is rising, in terms of how fast the head is rising.

### BEHIND THE BUTTON 135W

What the last operator's trace shows. Motor cut at the mark, trolley eighteen metres past it. That is not a mistake about where the mark was; it is a mistake about what cutting power does. On a level rail with little friction, removing the drive leaves the speed almost unchanged.

Why braking has to start early. The distance needed to stop depends on the speed and on how hard you can brake. Both are known before the run, so the braking point is calculable rather than something to be judged as the mark approaches.

Why a smaller pulse is easier to fly. A gentle start takes longer and asks less of the braking, so an error in the braking point costs less. Speed is not being scored here; arriving at the mark at rest is.

### TAKES AS READ

the discharge over a weir depends on the head raised to a power · a quantity that changes because another one changes needs the chain rule

### VERDICT 94W

The discharge does not follow the gate; it follows the head, and it follows it through a three-halves power. That exponent is why a spillway is hard to run: the same half metre of head does far more when the head is already high, so a rate that looked manageable an hour ago is not the rate you have now. The chain rule is what carries the head's own motion into the discharge, and dropping its inner factor leaves you with  $dQ/dH$  — a slope that is perfectly correct about a question nobody asked.

### TAKEAWAY 16W

A rate on the left needs every factor the right-hand side's own change brings with it.

### END OF THE DAY 15W

A route learned as places can be rejoined from anywhere in it; a count cannot.

## What speed does to a decision

### BEFORE THE DAY — EVADE

#### The client who wants the result before the check

He has been at the gatehouse since eight wanting the number before the second analyst has seen it, and the whole point of this office is that nothing leaves on one pair of eyes. Keep clear until the check is signed.

### PLAN CARD 148W

Friday, and the quarterly verification exercise runs all day. Every task on it is one somebody here does under time pressure in an ordinary week: the error notes sorted before the courier goes, the calibration bath held while a reference cell settles, the certificate board whose standing instruction changes the moment the harbour job lands. Priya Raman is running it and has said plainly that nobody is being tested on how fast they are. Today you sort the fortnight's errors, drive the sign-off round, hold the bath, clear the board and get a line across the river — and what the exercise measures is whether the category, the band and the instruction are still being consulted when there is no time left to reason them out. An office that only checks carefully when it is not busy is an office that checks nothing on the days it matters.

### BRIEFING 21W

A sorting line, a lap of the site, a bath to hold, a board to clear, and a line to throw.

### WHAT YOU DECIDE 14W

Meet the five panels where the pressure is the clock rather than the reasoning.

## 8.1 Which kind of error is it

FIELD STATION · A ROOM · BELT

### SCENE 29W

The fortnight's error notes have to go to the courier in ten minutes, sorted. One pile can be corrected after the fact. The other can only be averaged down.

### GUIDE 58W

Two bins. A systematic error pushes every reading the same way, so it can be found and taken out afterwards. A random error scatters readings either side of the truth, so it cannot be removed from a single reading and shrinks only when you take more of them. Sort on that, not on how the note is worded.

### ASK 14W

Sort each note into the pile that says how it can be dealt with.

### BEHIND THE BUTTON 102W

Why the two piles are treated so differently. A systematic error has a sign. Find it and every affected reading can be corrected by the same amount, years later if necessary. A random error has no sign to find, so the only remedy is more readings — and the scatter falls as the square root of how many.

Why the wording will not help you. Both piles contain drifts, both contain noises and both contain things that sound mechanical. The word is not the category, and a belt that could be sorted by the word would be measuring whether you can read.

### TAKES AS READ

a measurement can be wrong in a way that repeats, or wrong in a way that scatters

### VERDICT 145W

The two piles are not degrees of the same thing. A systematic error has a direction: a thermal drift, a stale calibration, a scale that reads 0.4 high all afternoon. Once it is identified every reading it touched can be corrected by the same amount, which is why old data can be rescued years later. A random error has no direction to find. Vibration from traffic, a flickering last digit, the operator's timing — each one moves a reading either way, so no correction can be applied to a single number and the only remedy is to take more of them and let the scatter fall as the square root of the count. That is the whole of why the courier gets two piles rather than one list. And the wording is deliberately no help: there are drifts on both sides and noises on both.

### TAKEAWAY 17W

A systematic error can be corrected after the fact; a random one can only be averaged down.

## 8.2 The round nobody can shorten

RECORDS OFFICE · A ROOM · TRIAL

### SCENE 30W

The quarterly round has five stops and one van. The route that covers the least ground is not the route the paperwork allows, and Wednesday's driver took the short one.

### GUIDE 56W

Drive the five gates in the order the work has to happen, not the order that covers the least ground. The sample cannot be signed in before it is collected, and it cannot be collected before the station is opened for the day. The clock is running and it is not what you are graded on.

### ASK 12W

Drive the five gates in the order the round has to happen.

### BEHIND THE BUTTON 98W

Why the shortest route is the wrong one. Each stop hands something to the next. Driving them in the order that saves fuel means arriving somewhere with nothing to do there, going back for it, and signing a form out of sequence — which is what Wednesday's driver did and why the batch had to be repeated.

What the clock is for. Pressure, and nothing else. The grade is the order. A lap driven carefully in the right order beats a fast one that saves the records office until after the desk has already been asked to respond.

### TAKES AS READ

a round of stops can have an order that its paperwork requires

### VERDICT 114W

The round is a chain of handovers. The gatehouse opens the day and issues the paperwork; nothing legitimately starts before it. The field station is where the sample is taken, and the sample room cannot receive what has not been collected. Records cannot file a result that the sample room has not produced, and the response desk cannot act on a finding that is not on the record. Driven in that order every stop has something waiting for it. Driven in the order that covers the least ground — station, gatehouse, records, desk, sample room — two stops are visited before the thing they need exists, which is exactly the lap Wednesday's driver drove.

### TAKEAWAY 15W

The route that covers the least ground is not always the route the work allows.

### 8.3 Twenty degrees, for three quarters of a minute

SAMPLE ROOM · A ROOM · HOLD

#### SCENE 36W

The reference cell needs the calibration bath at twenty degrees while it settles. The door to the corridor is propped, a heater on the next bench cycles, and the fan comes on when the room warms.

#### GUIDE 54W

Hold the bath at twenty degrees. The band is drawn on the dial and it narrows as the cell settles, because a cell that has nearly settled is disturbed by less. The trim wheel is the only control you have, and finding out which way it moves the temperature is part of the job.

#### ASK 10W

Hold the bath inside the band while the cell settles.

#### BEHIND THE BUTTON 95W

Why the band narrows. Early on the cell is still coming to temperature and a tenth of a degree either way changes nothing. As it settles the same tenth of a degree becomes the difference between a usable calibration and one that has to be repeated.

What is actually pushing it. Three things, at known times: the corridor door, the heater on the next bench cycling, and the fan. None of them is large. All of them are steps in the RATE, so a bath left alone does not sit still — it keeps going.

#### TAKES AS READ

a control can move a quantity up or down, and takes time to do it

#### VERDICT 126W

Each disturbance here is a step in the rate rather than a step in the temperature. The corridor door does not drop the bath by three hundredths of a degree; it makes the bath lose three hundredths a second, and it goes on losing until something answers it. That is why a bath left alone leaves the band and keeps going, and why the trim wheel has to be moved to a new position rather than nudged and released. The second half of the job is the band narrowing. A correction that was comfortably good enough while the cell was coming to temperature is not good enough once it has nearly settled, so the same wheel has to be held closer as the run goes on.

#### TAKEAWAY 13W

A disturbance you do nothing about is not a disturbance that goes away.

## 8.4 The board changes its mind

RESPONSE DESK · A ROOM · SPOT

### SCENE 30W

Certificates arrive on the board all morning. The desk works to whatever the standing instruction says, and the instruction is changed by whoever is running the desk, without an announcement.

### GUIDE 50W

Take the certificates the instruction currently wants, and leave the rest. The instruction at the top of the board is replaced during the run and nobody will tell you when. What is scored is the seconds either side of each change, because that is the only interval that measures anything.

### ASK 10W

Clear the board to whatever the standing instruction currently says.

### BEHIND THE BUTTON 102W

Why the seconds after a change are the ones that count. Most of what arrives is wanted by neither instruction and is correctly ignored by somebody who has understood nothing. The items that separate a player who noticed from one who did not are the ones the old rule wanted and the new one does not.

Why this is not a reflex test. The instruction is on the screen the whole time. What costs you is continuing to work to the rule you had, which is a habit rather than a speed, and it is the thing a desk actually gets wrong.

### TAKES AS READ

a standing instruction can be replaced while work is still arriving

### VERDICT 121W

Three instructions run during the exercise and each one wants a different part of the board. Anything overdue, anything from the harbour site, anything not yet checked — and a certificate can be two of those at once, which is what makes the switch cost real rather than notional. What the panel scores is the window either side of each change, because everything else on the board is wanted by neither rule and is ignored correctly by somebody who has read nothing. The desk's own version of this is a Tuesday when the harbour job lands: the standing instruction changes, and the work that goes out wrong in the next quarter of an hour is the work somebody was already doing.

### TAKEAWAY 15W

The cost of a withdrawn instruction is paid by whoever is still working to it.

## 8.5 The line across the river

FIELD STATION · A PERSON · LOB

### SCENE 47W

The staff gauge is on the far bank and the bridge is four miles round. The line has to go across before the level can be read, and the throwing arm has an angle and a charge and no readout of what it is about to do.

### GUIDE 49W

Set the angle and the charge, and throw. Three marks, working outward. The arm does not tell you the speed it will give the line, so the only way to find it is to throw and watch — which is what the second shot at each mark is for.

### ASK 9W

Land the messenger line on each mark, working outward.

### BEHIND THE BUTTON 107W

Why there is no number to work from. The arm's spring is old and its charge dial is relative, so nobody can tell you the launch speed. Everything else in this office is a measurement; this is a knack, and pretending otherwise would put a calculation in front of something that is genuinely learned by throwing.

What the shape of the flight does tell you. A steeper throw spends longer in the air and lands shorter, a shallower one flies flatter and further, and past a point the extra charge is spent on height rather than distance. Two throws at a mark are enough to bracket it.

### TAKES AS READ

a thrown object rises and falls, and both angle and effort change where it lands

### VERDICT 120W

Three marks at three distances, and the arm gives you no number to start from. What the flight gives you instead is a bracket: a throw that lands short and a throw that lands long put the mark between two settings you have actually seen, and the third throw goes between them. That is a real method, and it is the one used when a quantity cannot be read off anything. The far ladder is the mark that teaches the rest of it. Aiming higher for a longer throw stops working past forty-five degrees, because the extra charge goes into height rather than distance — so the ladder wants a flatter throw than the bollard did, not a harder one.

### TAKEAWAY 12W

Some quantities are measured, and a few are learned by trying them.

### END OF THE DAY 19W

A decision made at speed is still a decision, and what it costs is the thing you stop checking.



## Where the answer is the place

### BEFORE THE DAY — TAG

#### Catch Oyelaran before the balance room is closed

Oyelaran, the sample room lead, has the morning's drift check for the balance, and the room is locked at noon for the air conditioning work. Every mass this afternoon is corrected against that check, and without it the day's weighings are numbers with no traceability.

### PLAN CARD 172W

Thursday, and the annual audit lands on Monday. None of what has to happen today can be done sitting down: the fortnight's loggers are still out where the storm left them, the auditor wants to walk the sampling route with somebody who can answer for it, and Ada Merrick wants a straight answer on whether the new duplicate procedure has actually changed anything before she signs a page saying it has. Tobias Lindqvist has the list of who has to be told about Monday, and telling somebody is walking to where they are standing. Today you get round the station, walk the route with Merrick, recover six of the loggers, ask the site whether the procedure has changed anything, keep the open tray clear while it settles and catch the analyst Petar Ilić before an unsigned form goes out of the gate. What all six have in common is that the work is somewhere else, the hour is real, and the order you choose is the only part of it you control.

### BRIEFING 22W

A round of the station, a walk to keep up with, a search, a straw poll, a clear radius and a chase.

### WHAT YOU DECIDE 15W

Meet the six panels that are graded against the site rather than against a board.

## 9.1 Getting round them all

FIELD STATION · A ROOM · GREET

### SCENE 31W

Monday's audit has to be told to everyone on site, in person, before the morning is out. Fourteen people work here, spread across four buildings, and none of them stays put.

### GUIDE 50W

Ten of the fourteen have to hear it from you. Get to somebody and press Use to tell them; the rings on the ground are the people still to hear it. Which ten you reach is settled before you set off, by the order you choose to walk them in.

### ASK 12W

Tell ten of the fourteen about Monday before the hour is out.

### BEHIND THE BUTTON 87W

Why ten of fourteen rather than all of them. A round with no choice in it is a route somebody else planned. Four buildings hold the fourteen unevenly, so the question is which cluster to spend the walk on and which to leave.

Why an hour. At walking pace the four buildings are about forty minutes apart in the best order and considerably more in the worst, so the hour is tight enough that the order decides the outcome and loose enough that a sensible one arrives.

### TAKES AS READ

a place has a shape, and walking it costs time in proportion

### VERDICT 117W

The people are not evenly spread and that is the whole of it. Four of them stand at the field station, four at the sample room, three at records and three at the desk, so the cost of the round is almost entirely the walking between buildings rather than the talking. A route that takes the two nearest clusters and then the third reaches ten inside the hour; one that crosses the site twice does not, and the four people you never reach are decided by a choice you made in the first two minutes. That is what a round is, and it is why the list is longer than the number that has to hear it.

### TAKEAWAY 21W

A round is a route, and who you reach is decided by the order before it is decided by the walking.

## 9.2 Staying with her

FIELD STATION · A PERSON · FOLLOW

### SCENE 31W

Merrick walks the sampling route once a fortnight and will not wait for anybody. She stops at each point long enough to look, and says nothing when she moves off again.

### GUIDE 47W

Stay between three and fourteen metres of her for the whole walk. Closer than three and you are in front of her the moment she stops; further than fourteen and you are guessing which way she went at the last corner. She pauses at each sampling point.

### ASK 11W

Stay with her from the station to the last sampling point.

### BEHIND THE BUTTON 82W

Why there is a near limit at all. Following somebody is not chasing them. A person who is being walked into stops explaining what they are looking at, and somebody who is ahead of the walk is no longer being shown anything.

Why the pauses matter more than the pace. She walks slower than you can, so nothing here is a race. The failures all happen where she stops or turns, which is exactly where somebody treating it as a race overshoots.

### TAKES AS READ

a distance can be too small as well as too large

### VERDICT 124W

A band has two edges and only one of them feels like failure. Dropping behind is obvious: you lose sight of her at a corner and have to guess. Getting too close is not, because it feels like doing the task well, and it is the one that ruins the walk — she stops at each point to look at something, and somebody four paces from her shoulder arrives in front of the thing being looked at. She walks at about two thirds of your pace, so nothing here rewards speed. What it rewards is noticing that she has stopped before you have walked past her, which is the whole difference between being shown a route and merely arriving at the end of one.

### TAKEAWAY 19W

Following is a band rather than a line, and both edges of it are a way of losing somebody.

### 9.3 Six of the ten, before the time goes

SAMPLE ROOM · A ROOM · HUNT

#### SCENE 36W

Ten data loggers went out before the storm and none of them came back on their own. The sample room needs six of them working by Monday. Nobody wrote down where any of them ended up.

#### GUIDE 57W

Six of the ten are enough, and they are not on the map. They are small orange cases lying where the water left them — against a wall, off the edge of a path, behind something. You have to walk the ground to see one, so the order you sweep it in is the whole of it.

#### ASK 12W

Bring back six of the ten loggers before the time is out.

#### BEHIND THE BUTTON 99W

Why a sweep beats wandering. Ground you have already walked is worth nothing to walk again, and the only way to know which ground that is, is to have covered it in some order. A search with no shape to it re-walks its own middle.

Why six rather than ten. Four of them are far enough out that fetching them costs more than the other six put together, and nobody knows in advance which four those are. A stopping rule is part of a search, and a search with no stopping rule is just walking until the time goes.

#### TAKES AS READ

a route has a cost, and the cost is not the same for every order

#### VERDICT 117W

Nothing here tells you where anything is, which is what makes it a search rather than a delivery round. Ten cases are lying about the site and six will do, so the question is never which one to fetch next — it is which ground to cover, in what order, and when to stop covering it. A sweep that works outward from the sample room sees new ground with every step. A player who crosses the site to a place they remember something being re-walks the middle twice and arrives with four. The far cases are not worth more than the near ones; they are only further away, and the clock does not care which is which.

#### TAKEAWAY 20W

A search is ground covered in an order, and the order is what stops you walking the same yard twice.

## 9.4 How many is enough

RECORDS OFFICE · A ROOM · CANVASS

### SCENE 36W

The new duplicate procedure was piloted at the field station in March and rolled out everywhere in May. Merrick signs a page on Monday saying it has cut disagreements between runs. Nobody has asked the site.

### GUIDE 47W

Get to somebody, press Use, and they will tell you whether the procedure has made a difference in their own work. Everybody answers. Nobody will tell you when you have asked enough, and the answer you report at the end is your own — not the tally.

### ASK 12W

Has the new duplicate procedure cut disagreements between runs across the site?

### BEHIND THE BUTTON 97W

Why the sample is the question. Fourteen people is small enough to ask all of them and large enough that you will not, and the place is laid out so that the people nearest the office are not a fair picture of it.

Where the pilot ran. The field station has been using the procedure since March and everybody else since May, so the two groups have had very different amounts of time with it. That is a real difference between the people you can ask, and it is the reason the nearest answers lean one way.

### TAKES AS READ

a sample can be unrepresentative without anybody lying

### VERDICT 119W

Four of the five people nearest this office work at the field station, where the procedure was piloted in March, and they will tell you it works. They are not wrong about their own work. They are four people out of fourteen, and the other ten have had the procedure for a few weeks rather than a few months. Ask the near cluster and stop, and you have measured where you were standing rather than the thing you asked about — and the sample will be internally consistent while you do it, which is why nothing about the answer feels wrong at the time. The remedy is not a bigger sample. It is a sample that crosses the site.

### TAKEAWAY 18W

A sample is not a smaller copy of a place; asking the nearest few measures your own route.

## 9.5 Keeping the radius

RESPONSE DESK · A ROOM · EVADE

### SCENE 40W

The tray is open and has to settle for half a minute before it can be closed. Nobody may come within nine metres of it while it does. Ruth Calderón wants a signature and is following you to get one.

### GUIDE 46W

Hold nine metres of clear ground for thirty seconds. The count runs only while you are clear and stops while she is closer than that, so being caught costs seconds rather than the drill. She walks slower than you and she does not walk through buildings.

### ASK 13W

Hold nine metres of clear ground for thirty seconds while the tray settles.

### BEHIND THE BUTTON 94W

Why the count stops rather than resets. A drill that ends the moment somebody gets near it is measuring reaction time. One that costs you the seconds is measuring whether you can keep hold of a constraint while somebody is trying to talk to you.

Why nobody is faster than you. If she were, the only outcome would be decided by how fast you can press a key. She is not, so the drill is decided by where you go — which means the buildings, the gaps between them and the fence behind you.

### TAKES AS READ

a fixed distance can be maintained by choosing ground rather than by speed

### VERDICT 122W

Nobody in this drill is faster than you, so being caught is never about speed. It is about the ground you were standing on when she arrived. Walk in a straight line and the fence at the edge of the site ends the run for you — there is nowhere further to go and she closes the last few metres while you stand at it. Put a building between you and she has to walk round it, which buys the seconds the tray needs. The count stopping rather than resetting is deliberate: the cost of losing the radius is the time it takes to get it back, which is exactly what it costs in the room this drill is standing in for.

### TAKEAWAY 20W

A radius held under pressure is a matter of ground, not of speed, once nobody involved is faster than you.

## 9.6 Catching him before the gate

SAMPLE ROOM • A PERSON • TAG

### SCENE 35W

Ilić is walking the afternoon tray down to the courier van and the custody form on it has not been signed. Once it is through the gate it is somebody else's problem for a fortnight.

### GUIDE 50W

Get within two and a half metres of him inside half a minute. He walks away from you and nowhere else, and he is slower than you are — so this is not a race. Walking straight at him closes the gap slowly enough that he reaches the gate first.

### ASK 9W

Catch Ilić before the tray goes through the gate.

### BEHIND THE BUTTON 90W

Why a straight line does not work. Two people walking in the same direction close at the difference between their paces, not the sum. Against somebody two thirds your speed that difference is small, and half a minute of it is not enough ground.

What does work. He turns away from wherever you are, so where you stand decides where he goes. Come at him from the side he wants and the fence, the sample room wall and the corner of the yard all do the work the walking cannot.

### TAKES AS READ

two people moving the same way close at the difference between their speeds

### VERDICT 111W

The arithmetic is the whole of it. He walks at about two thirds of your pace, so following directly behind him closes the distance at the difference between the two — a metre and a half a second against fifty metres of yard, which is more than the half minute you have. Nothing about that improves if you try harder. What changes it is that he moves away from wherever you are standing, so your position steers him: come round one side and he turns toward the fence, and the ground he has left runs out before your time does. That is a chase decided by geometry rather than by pace.

### TAKEAWAY 22W

A chase against somebody slower is won by the line you take, because closing speed is a difference and not a sum.

### END OF THE DAY 24W

Work that happens across a place is planned as a route, and the route is a decision that is made before the walking starts.

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